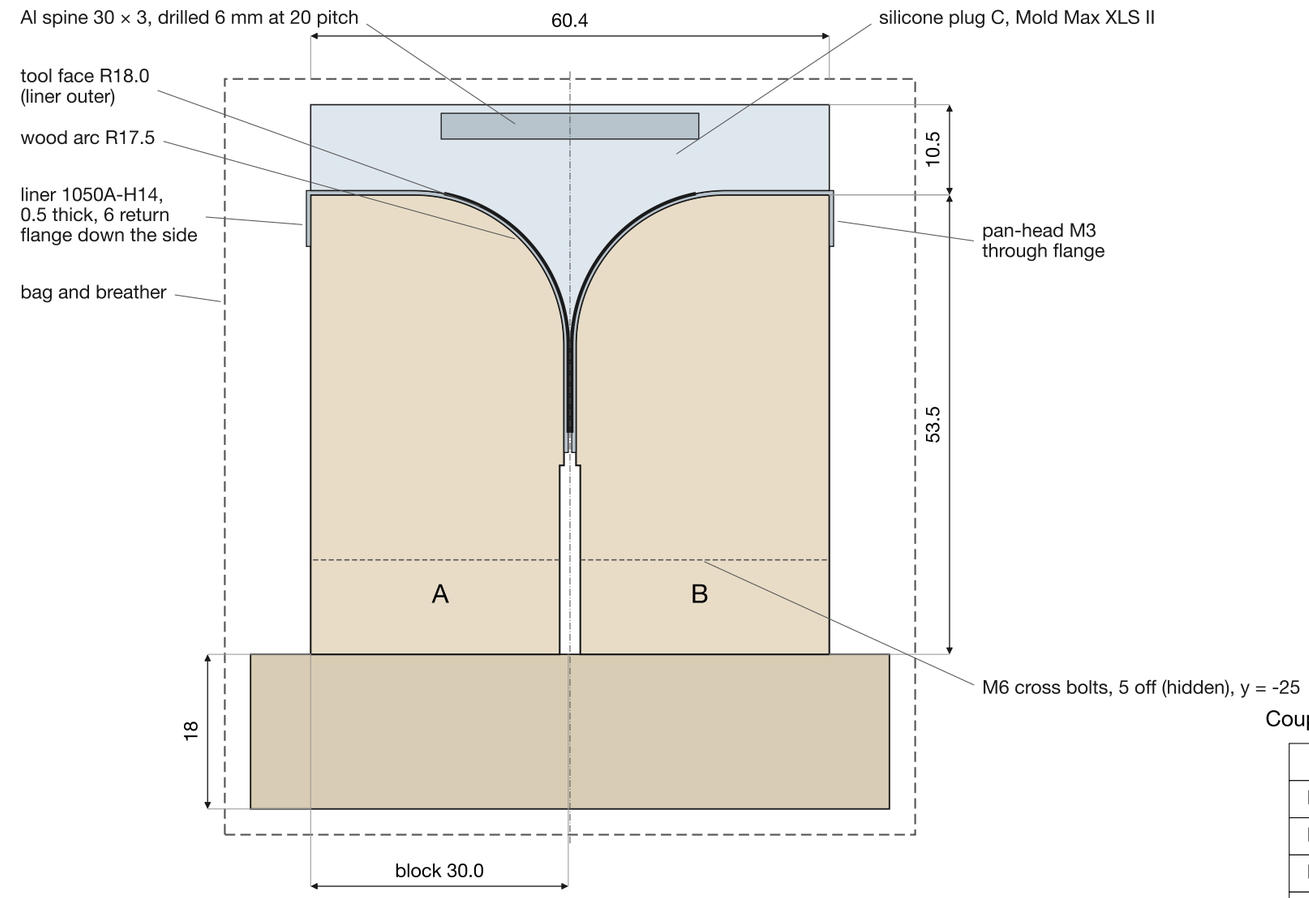
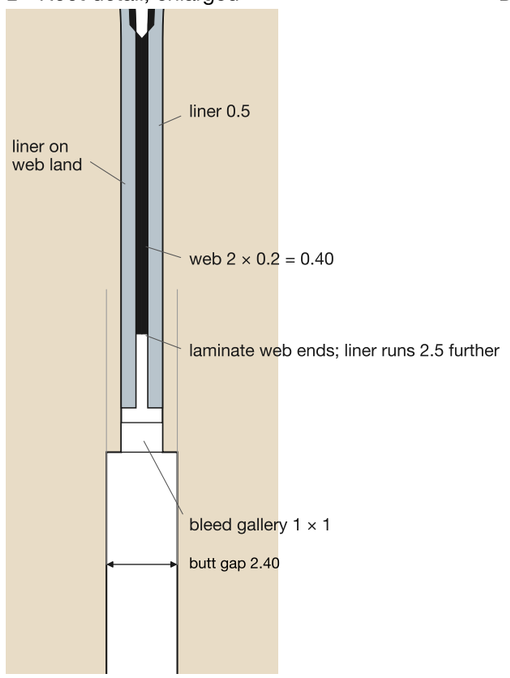
A Section through the closed tool, mid-length

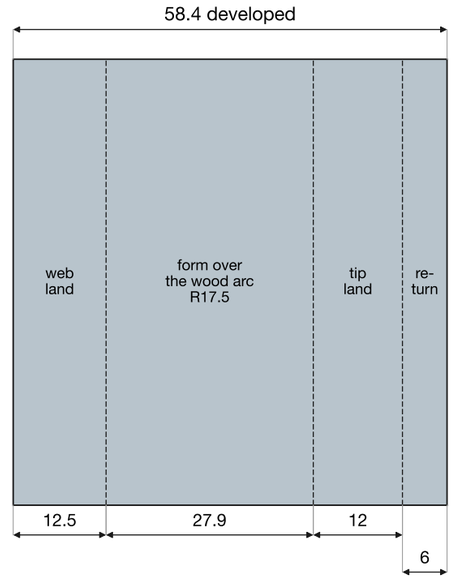


B Root detail, enlarged



The 0.36 stops sit between the liner faces, at the trim ends only (see C).

D Liner blank, flat, 2 per tool

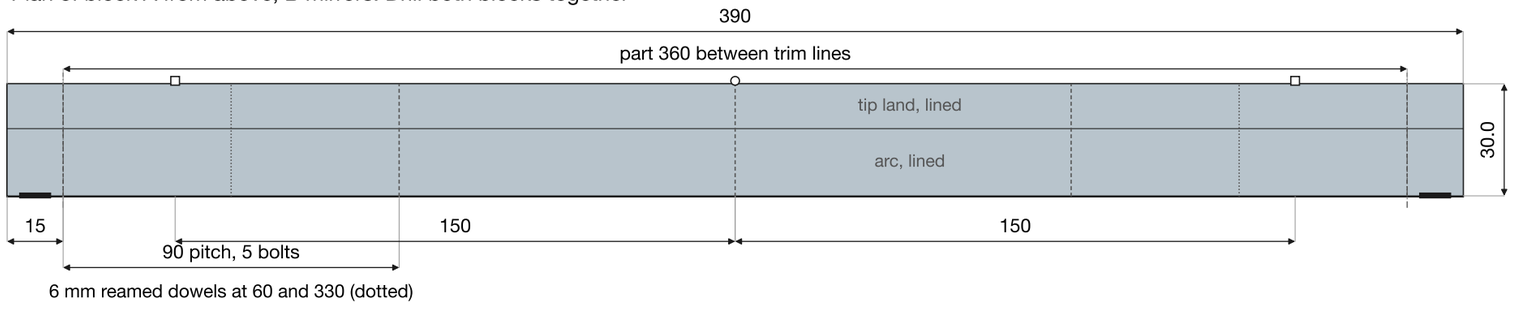


length 390; 60 shown. Pre-form over a R14.8 former, fit cold on the block to a 0.1 feeler, trim ends flush.

Coupon blocks, 5 pairs, 60 long, all in one cure

tool face	wood arc	what it tests
R15.2	R14.7	spring-in by Radford's thick-laminate formula (calculated)
R16.5	R16.0	halfway between rows 1 and 3
R18.0	R17.5	spring-in as Leclerc measured (part R = 0.835 × tool R, test data)
R19.6	R19.1	spring-in as a fixed change in curvature
R18.0	R18.0, bare	same as row 3, no liner: does the liner change the part?

C Plan of block A from above; B mirrors. Drill both blocks together



M3 pan head in the return flange, outer face: round hole at mid-length (circle), 3 mm slots either side (squares)

0.36 stops (black), web land, outside the trim lines

NOTES

1 Wood: one 35 mm phenolic birch strip per block, no glueline. Rough, bake to constant mass, finish, seal.
2 Liner: 1050A-H14, 0.5 mm, micrometer at 3 points (reject spread > 0.02). Not bonded. FEP under it on the web lands, Frekote elsewhere. Measure the friction coefficient (μ) in a 130 °C pull test first; if it is above 0.15 use 5754-H22.
3 Long blocks: rough the arc only until the coupons are read. Finish all arcs on CNC; no stock cutter matches.
4 Plug C: cast on the closed, lined pair over a measured 0.22-0.25 mm spacer. 24 h cure at room temperature.
5 FEP on the tip lands, and under the liner on the web lands. Nothing across the V.
6 Tolerances: web lands flat 0.05 over 390 and parallel 0.05; arc radius ±0.05; root tangent, no step.

Svarog TRAC boom mould, v5

Thomas van Pul | ICSS Project Svarog | 2 Nov 2026

Dimensions in mm, do not scale | A3 | sheet 1 of 1 | supersedes v4

Svarog TRAC boom mould, version 5

Thomas van Pul · ICSS Project Svarog, structures · 2 November 2026 · Drawing: mould_v5.pdf · Numbers: calc5.py

Terms. TRAC: triangular rollable and collapsible boom. Web: the flat wall where the two halves of the boom are bonded. Flanges: the two curved halves either side of it. Land: a flat strip on a block. Liner: the 0.5 mm aluminium sheet on each block's forming face. Spring-in: how much tighter the cured part curls than the tool it was made on.

1 Summary

Version 5 keeps the version 4 tool. Two plywood blocks clamp the web, a cast silicone plug presses both flanges, and the whole assembly is bagged and cured in the autoclave. The change is a 0.5 mm aluminium liner over each block's forming face, and a coupon run that settles two questions at once: which radius to cut, and whether the liner stays. Nothing has been cut yet. Whether the aluminium sheet has been ordered is still open (section 4).

2 What changed from version 4

Item	v4	v5	Why
Forming face	sealed plywood	0.5 mm 1050A-H14 liner, not bonded	smooth, non-porous face over machined end grain
Wood arc radius	R18.0	R17.5 (tool face still R18.0)	makes room for the liner
Wood web land	10 mm	13 mm	liner runs 2.5 mm past the end of the laminate web
Gap between the blocks' butt faces	1.4 mm	2.4 mm	leaves room for the liner on each web face
Cross bolts, depth below the web	23.5 mm	25 mm	stay central in the solid wood under the longer land
Liner fixing	none	6 mm return flange down the outer face, pan-head M3	no screw on a forming face or under the plug
Coupons	4 pairs	5 pairs, one bare at R18.0	shows whether the liner changes the part

3 Why the aluminium sheet goes back on

To say at the session

We're putting the half-millimetre aluminium sheet back on the blocks. Version 4 dropped it for two reasons. First, a vacuum bag couldn't press a badly formed sheet flat. Second, glued to the wood it would tear off, because over the cure it grows nearly a millimetre more than the plywood. This time we bend it over a tighter former first, so it already fits the block, and we don't glue it: one screw in the middle, two in slots on the side, release film underneath, so it slides every cure. What we gain is that the laminate cures against smooth metal, and the wood's rough end grain and its moisture stay behind the sheet. One of the five test coupons runs without the sheet, and if that one comes out as good, we take the sheet off the long tool.

4 Still to find out

Question	Ask	Why it matters
Cure temperature of our prepreg	lab technician, prepreg datasheet	the wood tool is limited to about 130 °C
Does the lab vent the bag during the autoclave cycle?	lab technician	sets the real pressure on the web
Was the 1050A sheet ordered, and in which temper?	Sahil	H14 is needed; O will yield and wrinkle

5 Liner numbers

Quantity	Value	Note
Sheet	1050A-H14, 0.5 ± 0.04 mm	measure at 3 points; reject a spread over 0.02
Flat blank, per block	58.4 × 390 mm	12.5 web land, 27.9 arc, 12 tip land, 6 return
Springback if bent on the block itself	R17.5 → R21.4	edge stands about 4 mm proud
Former radius to bend it over	R14.8	springs back to R17.5; open by hand if tight
Bag-only gap: vacuum, 3 bar, 7 bar	0.72 / 0.16 / 0.08 mm	why the bag is not trusted to seat it
Growth over plywood, 130 / 150 °C cure	0.75 / 0.89 mm	outer screws move 0.29 / 0.34 mm in 3 mm slots
Friction vs yield, 7 bar, released (μ 0.1)	1 324 N vs 2 226 N	slides, stays elastic
Friction vs yield, 7 bar, dry (μ 0.3)	3 972 N vs 2 226 N	yields: release under the liner is required
Effect on the 0.40 mm web gap	± 0.08 mm	stops set from the micrometer readings
Effect on the tool radius	± 0.2 %	negligible next to the ± 13 % uncertainty in spring-in

6 Order of work

1. Measure the sheet thickness with a micrometer at 3 points.
2. Friction test: pull a 20 mm strip of sheet from between two sealed, release-coated ply faces at 130 °C and record the friction coefficient. If it is above 0.15, switch to 5754-H22.
3. Rough-cut the long blocks and the ten coupon blocks from 35 mm phenolic birch. Bake to constant mass.
4. Finish the coupon arcs on the CNC (wood radii 14.7, 16.0, 17.5, 19.1, and 18.0 bare). Leave the long blocks rough. Seal.
5. CNC an R14.8 former from ply offcut. Bend the coupon liners over it, fit each to its block until a 0.1 mm feeler will not enter, screw through the return flange.
6. Set the 0.36 mm stops between liner faces from the micrometer readings.
7. Cast the plugs over a measured 0.22 to 0.25 mm spacer. Leave 24 hours.
8. Cure all five coupon pairs in one autoclave run. Measure radius and angle at mid-length; section each root.
9. Pick the long-tool radius from the coupons, and keep or drop the liner from the bare pair. Then finish the long blocks.

7 Parts new in version 5

Part	Quantity	Note
1050A-H14 sheet, 0.5 mm	one 500 × 500 mm	2 long liners and 8 coupon liners. O temper will not do.
FEP release film, 0.05 mm	1 m × 50 mm strip	under the liner on the web lands only
Frekote 700-NC	1	both liner faces, sealed wood
M3 × 12 pan-head wood screws, stainless	30	3 per block, long and coupon
18 mm ply offcut, 400 × 60	1	for the R14.8 former
5754-H22 sheet, 0.5 mm	only if needed	if the pull test reads above 0.15

Everything above comes from calc5.py (output in calc5_output.txt), which builds on calc4.py. The engineering review of this version and what changed because of it is in review-v5-adversarial.md.

Svarog structures session

Monday 2 November 2026 · Thomas van Pul · with the new members

1 From Sahil this afternoon

Topic	What he said
Structure changes	An M4 heat-set insert, and an M8 bolt on top to lower the centre-of-mass height.
Structures status	Finalised on the structures side.
Initial design review	With the new members, next Monday or at their next session.
PDR	Stays on the second-last session of term.
Tonight	Ask the new members how confident they feel, and get them thinking about who the subteam lead will be.

Ask Sahil: which part each of the two changes sits on, and whether the M8 bolt adds mass we have not counted.

2 Questions for the new members

1. From 1 to 5, how confident do you feel about the boom and the mould so far? Which part is least clear?
2. What have you done before: CAD, CNC or hand machining, composites lay-up, testing?
3. Which piece would you like to own before the design review: the coupon blocks, the liner and former, the silicone plug, or the structure CAD?
4. Can you make next Monday for the initial design review, or is your next session better?
5. Subteam lead: who would want to be considered? No decision tonight; think about it before the review.

3 Mould, version 5, in two minutes

The tool is the version 4 tool: two plywood blocks, a silicone plug, bagged and cured in the autoclave. The aluminium sheet goes back on, bent over a tighter former first, held by one screw in the middle and two in slots, so it can slide. The coupon run settles the radius and whether the sheet stays. Drawing: mould_v5.pdf. Handout: pack_v5.pdf.

Needed from the team	Who
Cure temperature of the prepreg	_____
Does the lab vent the bag in the autoclave?	_____
Was the 1050A sheet ordered, and which temper?	Sahil
Who runs the 130 °C friction pull test?	_____

4 Notes